Week5 jenkins file code

```
pipeline {
    agent any

    stages {

        stage('Checkout') {
            steps {
                echo "Cloning Airway repository..."
                checkout scm
            }
        }

        stage('Test') {
            steps {
                echo "Checking Airway Python application"
                sh 'python3 -m py_compile app.py'
            }
        }

        stage('Run') {
            steps {
                echo "Checking Airway application code"
                  sh 'python3 -c "import ast; ast.parse(open(\'app.py\').read()); print(\'Airway application
code is valid\')"'
            }
        }

        stage('Package') {
            steps {
                echo "Packaging Airway project"
                sh 'tar -czf airway-project.tar.gz --exclude=.git --exclude=.venv .'
            }
        }

        stage('Archive') {
            steps {
                echo "Archiving Airway project"
                archiveArtifacts artifacts: 'airway-project.tar.gz'
            }
        }
    }

    post {
        success {
            echo 'Airway testing and packaging completed successfully!'
        }

        failure {
            echo 'Pipeline failed. Check Jenkins Console Output.'
        }
    }
}
```

Week6 commands

```
cd ~/Desktop/Airway
docker build -t airway-app:week6 .
docker run -d --name airway-week6-container -p 5007:5005 airway-app:week6
```

Week7 jenkinsfile

```
pipeline {
    agent any

    environment {
        PATH = "/Applications/Docker.app/Contents/Resources/bin:/usr/local/bin:/usr/bin:/bin:/usr/
sbin:/sbin"
    }

    stages {
        stage('Check Docker') {
            steps {
                sh 'which docker'
                sh 'docker --version'
            }
        }

        stage('Build') {
            steps {
                echo "Build Airway Docker Image"
                sh 'docker build -t airway-app .'
            }
        }

        stage('Run') {
            steps {
                echo "Run Airway Application in Docker Container"
                sh 'docker rm -f airway-jenkins-test || true'
                sh 'docker run -d -p 5006:5005 --name airway-jenkins-test airway-app'
            }
        }
    }

    post {
        success {
            echo 'Airway pipeline completed successfully!'
        }

        failure {
            echo 'Airway pipeline failed. Please check the logs.'
        }
    }
}
```

```
Week8 commands
docker build -t airway-app:week8 .
docker login
docker tag airway-app:week8 nuhakhan1305/airway-app:week8
docker push nuhakhan1305/airway-app:week8
kubectl create deployment airway --image=nuhakhan1305/airway-
app:week8
kubectl expose deployment airway --type=NodePort --port=5005
kubectl get svc airway
```

Week9 jenkins file

```
pipeline {
    agent any

    environment {
            PATH = "/Applications/Docker.app/Contents/Resources/bin:/opt/homebrew/bin:/usr/local/
bin:/usr/bin:/bin:/usr/sbin:/sbin"
    }

    stages {
        stage('Build Docker Image') {
            steps {
                sh 'docker build -t nuhakhan1305/airway-app:week9 .'
            }
        }

        stage('Docker Login') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'dockerhub-creds',
                    usernameVariable: 'DOCKER_USER',
                    passwordVariable: 'DOCKER_TOKEN'
                )]) {
                        sh 'echo "$DOCKER_TOKEN" | docker login -u "$DOCKER_USER" --password-
stdin'
                }
            }
        }

        stage('Push Docker Image to Docker Hub') {
            steps {
                sh 'docker push nuhakhan1305/airway-app:week9'
            }
        }

        stage('Deploy to Kubernetes') {
            steps {
                sh 'kubectl apply -f deployment.yaml'
                sh 'kubectl apply -f service.yaml'
            }
        }
    }

    post {
        success {
            echo 'Pipeline completed successfully!'
        }
        failure {
            echo 'Pipeline failed. Please check the logs.'
        }
    }
}
```